TOSHIBA Photocoupler GaAℓAs Ired & Photo−IC

TLP250

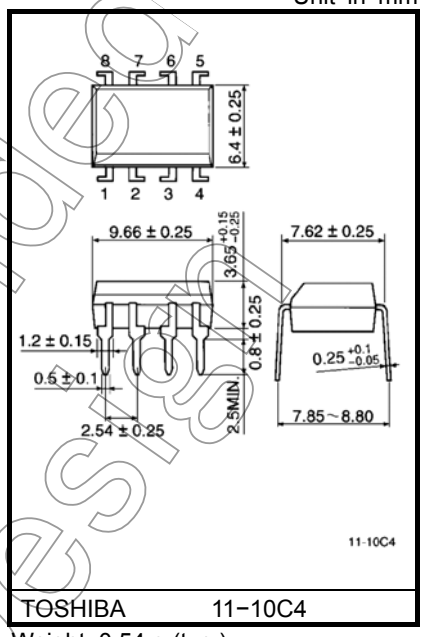
Transistor Inverter
Inverter For Air Conditioner
IGBT Gate Drive
Power MOS FET Gate Drive

The TOSHIBA TLP250 consists of a GaAlAs light emitting diode and a integrated photodetector.
This unit is 8−lead DIP package.
TLP250 is suitable for gate driving circuit of IGBT or power MOS FET.

- Input threshold current: I_F=5mA(max.)
- Supply current (I_{CC}): 11mA(max.)
- Supply voltage (V_{CC}): 10−35V
- Output current (I_O): ±1.5A (max.)
- Switching time (t_{pLH}/t_{pHL}): 0.5μs(max.)
- Isolation voltage: 2500V_{rms}(min.)
- UL recognized: UL1577, file No.E67349
- Option(D4)
  VDE Approved : DIN EN60747-5-2
  Maximum Operating Insulation Voltage : 890V_{PK}
  Highest Permissible Over Voltage : 4000V_{PK}

(Note):When a EN60747-5-2 approved type is needed, Please designate "Option(D4)"

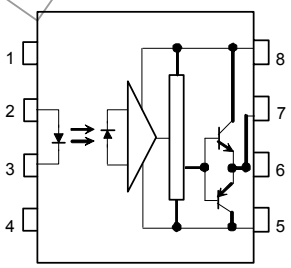
Unit in mm



TOSHIBA 11−10C4

Weight: 0.54 g (typ.)

Pin Configuration (top view)

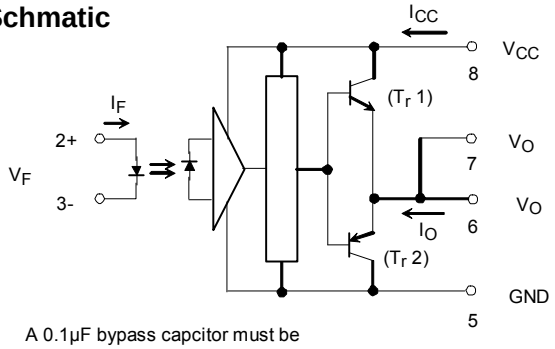


1 : N.C.
2 : Anode
3 : Cathode
4 : N.C.
5 : GND
6 : V_O (Output)
7 : V_O
8 : V_{CC}

Truth Table

		Tr1	Tr2
Input LED	On	On	Off
	Off	Off	On

Schmatic



A 0.1μF bypass capcitor must be connected between pin 8 and 5 (See Note 5).

Not Recommended for New Design

Absolute Maximum Ratings (Ta = 25°C)

Characteristic			Symbol	Rating	Unit
LED	Forward current		I_F	20	mA
	Forward current derating (Ta ≥ 70°C)		ΔI_F / ΔTa	−0.36	mA / °C
	Peak transient forward curent	(Note 1)	I_{FPT}	1	A
	Reverse voltage		V_R	5	V
	Junction temperature		Tj	125	°C
Detector	"H"peak output current (P_W ≤ 2.5μs,f ≤ 15kHz)	(Note 2)	I_{OPH}	−1.5	A
	"L"peak output current (P_W ≤ 2.5μs,f ≤ 15kHz)	(Note 2)	I_{OPL}	+1.5	A
	Output voltage	(Ta ≤ 70°C)	V_O	35	V
		(Ta = 85°C)		24	
	Supply voltage	(Ta ≤ 70°C)	V_{CC}	35	V
		(Ta = 85°C)		24	
	Output voltage derating (Ta ≥ 70°C)		ΔV_O / ΔTa	−0.73	V / °C
	Supply voltage derating (Ta ≥ 70°C)		ΔV_{CC} / ΔTa	−0.73	V / °C
	Junction temperature		Tj	125	°C
Operating frequency		(Note 3)	f	25	kHz
Operating temperature range			T_{opr}	−20~85	°C
Storage temperature range			T_{stg}	−55~125	°C
Lead soldering temperature (10 s)			T_{sol}	260	°C
Isolation voltage (AC, 1 min., R.H.≤ 60%)		(Note 4)	BV_S	2500	Vrms

Note: Using continuously under heavy loads (e.g. the application of high temperature/current/voltage and the significant change in temperature, etc.) may cause this product to decrease in the reliability significantly even if the operating conditions (i.e. operating temperature/current/voltage, etc.) are within the absolute maximum ratings.
Please design the appropriate reliability upon reviewing the Toshiba Semiconductor Reliability Handbook ("Handling Precautions"/"Derating Concept and Methods") and individual reliability data (i.e. reliability test report and estimated failure rate, etc).

Note 1: Pulse width P_W ≤ 1μs, 300pps

Note 2: Exporenential wavefom

Note 3: Exporenential wavefom, I_{OPH} ≤ −1.0A(≤ 2.5μs), I_{OPL} ≤ +1.0A(≤ 2.5μs)

Note 4: Device considerd a two terminal device: Pins 1, 2, 3 and 4 shorted together, and pins 5, 6, 7 and 8 shorted together.

Note 5: A ceramic capacitor(0.1μF) should be connected from pin 8 to pin 5 to stabilize the operation of the high gain linear amplifier. Failure to provide the bypassing may impair the switching proparty. The total lead length between capacitor and coupler should not exceed 1cm.

Recommended Operating Conditions

Characteristic	Symbol	Min	Typ.	Max	Unit
Input current, on (Note6)	$I_{F(ON)}$	7	8	10	mA
Input voltage, off	$V_{F(OFF)}$	0	—	0.8	V
Supply voltage	V_{CC}	15	—	30 / 20	V
Peak output current	I_{OPH}/I_{OPL}	—	—	±0.5	A
Operating temperature	T_{opr}	−20	25	70 / 85	°C

Note: Recommended operating conditions are given as a design guideline to obtain expected performance of the device. Additionally, each item is an independent guideline respectively. In developing designs using this product, please confirm specified characteristics shown in this document.

Note 6: Input signal rise time(fall time)<0.5μs.

Not Recommended for New Design

Electrical Characteristics (Ta = −20~70°C, unless otherwise specified)

Characteristic		Symbol	Test Cir−cuit	Test Condition		Min	Typ.*	Max	Unit
Input forward voltage		V_F	—	I_F = 10 mA , Ta = 25°C		—	1.6	1.8	V
Temperature coefficient of forward voltage		$\Delta V_F / \Delta Ta$	—	I_F = 10 mA		—	−2.0	—	mV / °C
Input reverse current		I_R	—	V_R = 5V, Ta = 25°C		—	—	10	μA
Input capacitance		C_T	—	V = 0 , f = 1MHz , Ta = 25°C		—	45	250	pF
Output current	"H" level	I_{OPH}	1	V_{CC} = 30V (*1)	I_F = 10 mA V_{8-6} = 4V	−0.5	−1.5	—	A
	"L" level	I_{OPL}	2		I_F = 0 V_{6-5} = 2.5V	0.5	2	—	
Output voltage	"H" level	V_{OH}	3	V_{CC1} = +15V, V_{EE1} = −15V R_L = 200Ω, I_F = 5mA		11	12.8	—	V
	"L" level	V_{OL}	4	V_{CC1} = +15V, V_{EE1} = −15V R_L = 200Ω, V_F = 0.8V		—	−14.2	−12.5	
Supply current	"H" level	I_{CCH}	—	V_{CC} = 30V, I_F = 10mA Ta = 25°C		—	7	—	mA
				V_{CC} = 30V, I_F = 10mA		—	—	11	
	"L" level	I_{CCL}	—	V_{CC} = 30V, I_F = 0mA Ta = 25°C		—	7.5	—	
				V_{CC} = 30V, I_F = 0mA		—	—	11	
Threshold input current	"Output L→H"	I_{FLH}	—	V_{CC1} = +15V, V_{EE1} = −15V R_L = 200Ω, V_O > 0V		—	1.2	5	mA
Threshold input voltage	"Output H→L"	V_{FHL}	—	V_{CC1} = +15V, V_{EE1} = −15V R_L = 200Ω, V_O < 0V		0.8	—	—	V
Supply voltage		V_{CC}	—			10	—	35	V
Capacitance (input−output)		C_S	—	V_S = 0 , f = 1MHz Ta = 25°C		—	1.0	2.0	pF
Resistance(input−output)		R_S	—	V_S = 500V , Ta = 25°C R.H.≤ 60%		1×10^{12}	10^{14}	—	Ω

* All typical values are at Ta = 25°C (*1): Duration of I_O time ≤ 50μs

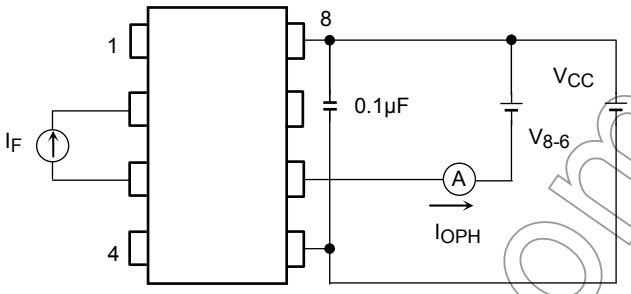
Not Recommended for New Design

Switching Characteristics (Ta = −20~70°C , unless otherwise specified)

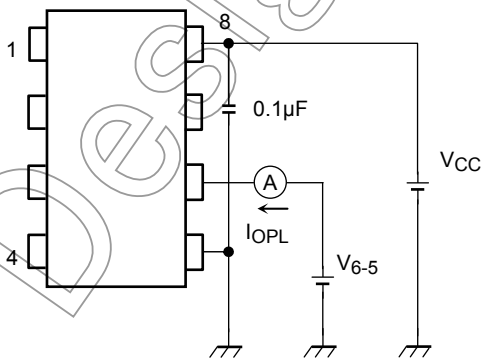
Characteristic		Symbol	Test Cir−cuit	Test Condition	Min	Typ.*	Max	Unit
Propagation delay time	L→H	t_{pLH}	5	I_F = 8mA V_{CC1} = +15V, V_{EE1} = −15V R_L = 200Ω	—	0.15	0.5	μs
	H→L	t_{pHL}			—	0.15	0.5	
Output rise time		t_r			—	—	—	
Output fall time		t_f			—	—	—	
Common mode transient immunity at high level output		C_{MH}	6	V_{CM} = 600V, I_F = 8mA V_{CC} = 30V, Ta = 25°C	−5000	—	—	V / μs
Common mode transient immunity at low level output		C_{ML}		V_{CM} = 600V, I_F = 0mA V_{CC} = 30V, Ta = 25°C	5000	—	—	V / μs

All typical values are at Ta = 25°C

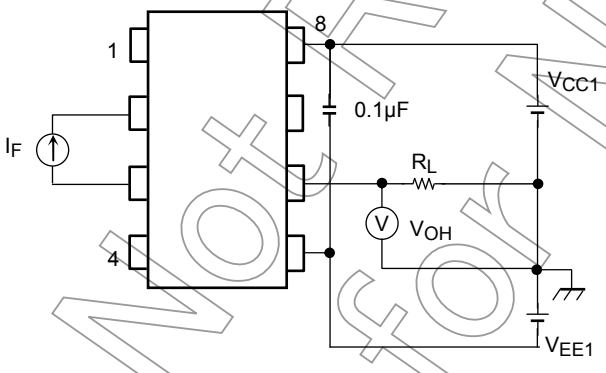
Test Circuit 1 : I_{OPH}



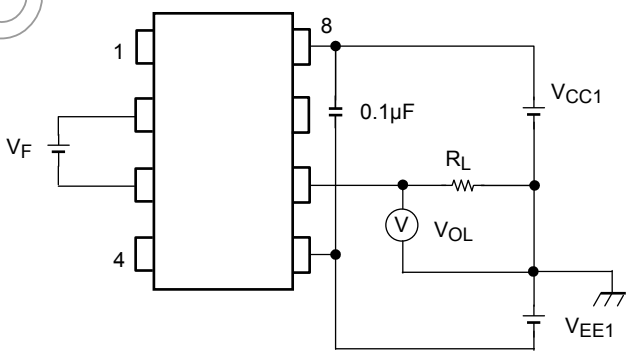
Test Circuit 2 : I_{OPL}



Test Circuit 3 : V_{OH}

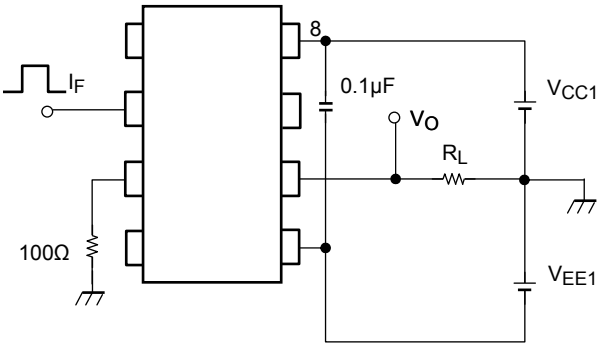


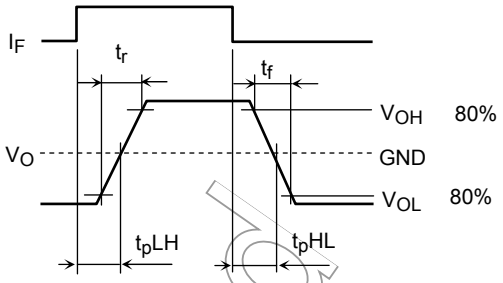
Test Circuit 4 : V_{OL}



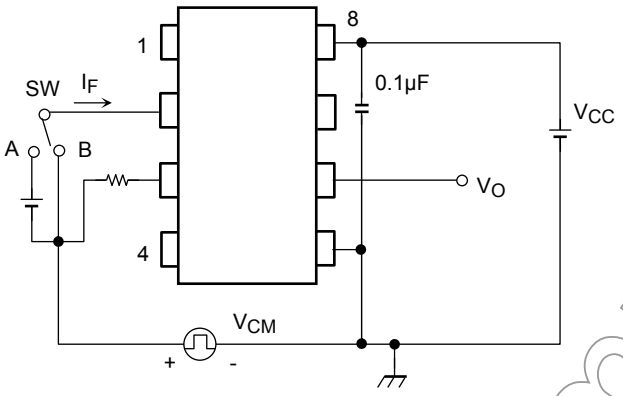
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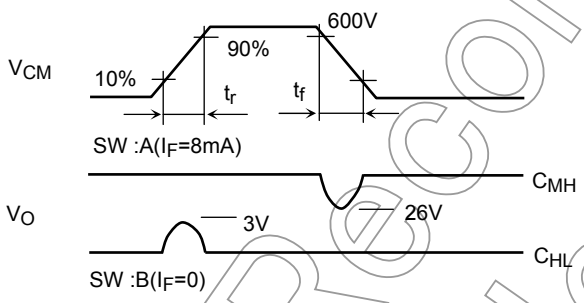
Test Circuit 5: t_{pLH}, t_{pHL}, t_r t_f





Test Circuit 6: C_{MH}, C_{ML}





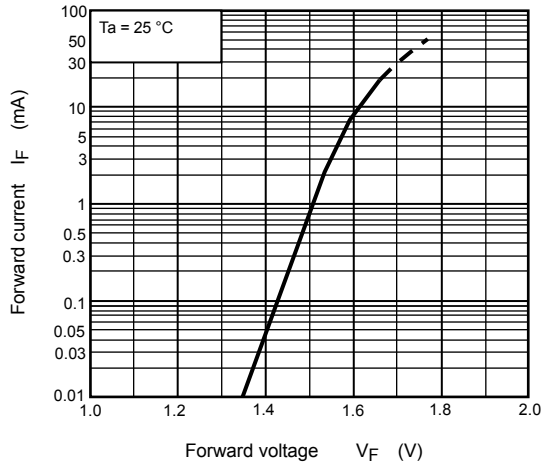
$$C_{ML} = \frac{480\ (V)}{t_{r\ (\mu s)}}$$

$$C_{MH} = \frac{480\ (V)}{t_{f\ (\mu s)}}$$

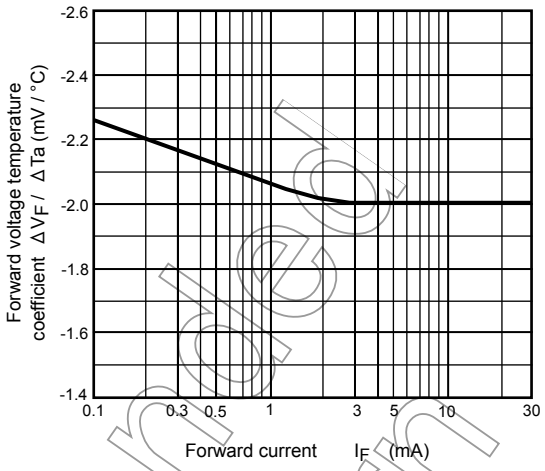
C_{ML}(C_{MH}) is the maximum rate of rise (fall) of the common mode voltage that can be sustained with the output voltage in the low (high) state.

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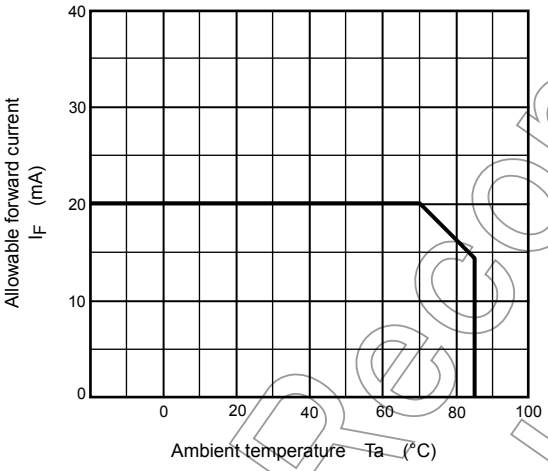
$I_F - V_F$



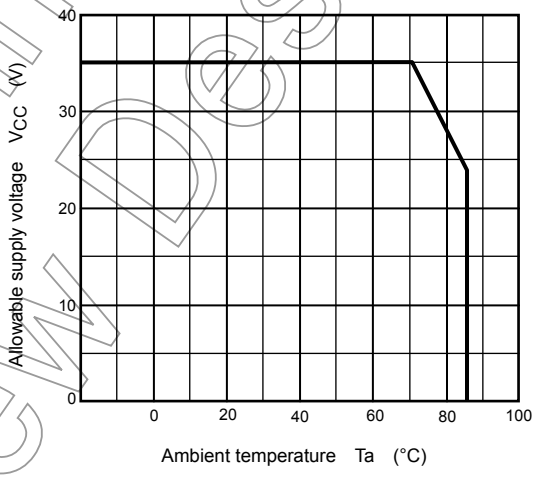
$\Delta V_F / \Delta Ta - I_F$



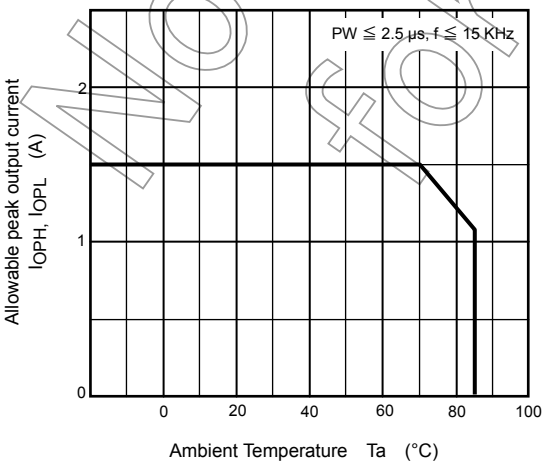
$I_F - Ta$



$V_{CC} - Ta$



$I_{OPH}, I_{OPL} - Ta$



Not Recommended for New Design

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